

04 – We can C more

Let's C more

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Source: several C code examples were generated using Mistral AI

OUTLINE

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1) DATATYPES, VARIABLES AND OPERATORS

```
1 // file lec-01.c
2 #include <stdio.h>
3
4 int main() { // the main function that is run automatically when the code is executed
5     int myInteger=123; // an integer (4 bytes)
6     char myChar='a'; // a character (1 byte)
7     float myFloat=3.14159265; // a real number (4 bytes)
8     double myDouble=3.14159265; // a real number (8 bytes)
9     char myString[100]="Hello, World!"; // this is a CAST
10    printf("My integer is: %d, it has a size of %d byte(s).\n", myInteger, (int) sizeof(int));
11    printf("My float is: %f, it has a size of %d byte(s).\n", myFloat, (int) sizeof(float));
12    printf("My double is: %f, it has a size of %d byte(s).\n", myDouble, (int) sizeof(double));
13    printf("My char is: %c, it has a size of %d byte(s).\n", myChar, (int) sizeof(char));
14    printf("My string is: %s, it has a size of %d byte(s).\n", myString, (int) sizeof(myString));
15    return 0; // 0 is returned at the end of the program
16 }
17
```

```
os2024@os2024-virtualbox:~/Documents/lecture-2024-10-23$ gcc lec-01.c && ./a.out
My integer is: 123, it has a size of 4 byte(s).
My float is: 3.141593, it has a size of 4 byte(s).
My double is: 3.141593, it has a size of 8 byte(s).
My char is: a, it has a size of 1 byte(s).
My string is: Hello, World!, it has a size of 100 byte(s).
os2024@os2024-virtualbox:~/Documents/lecture-2024-10-23$
```

1) DATATYPES, VARIABLES AND OPERATORS

```
1 #include <stdio.h>
2
3 int main() {
4     int a=123, b=234;
5
6     //a+=1;//a=a+1;//a++;
7     //b-=1;//b=b-1;//b--;
8     printf("a=%d\n", a);
9     printf("b=%d\n", b);
10    //b=a++;
11    //printf("b=%d\n", b);
12    b=++a;
13    printf("a=%d\n", a);
14    printf("b=%d\n", b);
15
16    return 0;
17 }
18
```

```
os2024@os2024-virtualbox:~/Documents/lecture-2024-10-23$ gcc lec-03.c && ./a.out
a=123
b=234
a=124
b=124
os2024@os2024-virtualbox:~/Documents/lecture-2024-10-23$
```

2) CONTROL STRUCTURES: IF

```

1 #include <stdio.h>
2
3 int main() {
4     int a=123, b=234;
5     int result=0;
6     float floatResult=0.0;
7
8     result=a+b;
9     printf("The result is: %d\n", result);
10    result=a-b;
11    printf("The result is: %d\n", result);
12    result=a*b;
13    printf("The result is: %d\n", result);
14    //floatResult=b*1.0/a*1.0;
15    floatResult=(float)b/(float)a; // This is a CAST
16    //result=(float)b/(float)a; // This is a CAST
17    printf("The result is: %f\n", floatResult);
18    //printf("The result is: %d\n", result);
19    int aNumber=12344;
20    if (aNumber%2==0) {
21        printf("The number is even.\n");
22    } else {
23        printf("The number is odd.\n");
24    }
25    aNumber=1232450;
26    if (aNumber%5==0) {
27        printf("The number is a multiple of 5.\n");
28    } else {
29        printf("The number is not multiple of 5.\n");
30    }
31    return 0;
32 }
33

```

```

os2024@os2024-virtualbox:~/Documents/lecture-2024-10-23$ gcc lec-02.c && ./a.out
The result is: 357
The result is: -111
The result is: 28782
The result is: 1.902439
The number is even.
The number is a multiple of 5.
os2024@os2024-virtualbox:~/Documents/lecture-2024-10-23$

```

2) CONTROL STRUCTURES: IF & TERNARY OPERATOR

```
1 #include <stdio.h> // include the library for printf()
2
3 // this is the main function
4 int main() {
5
6     int age=17; // age of a person
7
8     if (age>=18) {
9         printf("You are an adult!\n");
10    } else {
11        printf("You are not an adult!\n");
12    }
13
14    int a=11, b=3;
15    if (a>b) {
16        printf("The max value is: %d\n", a);
17    } else {
18        printf("The max value is: %d\n", b);
19    }
20    // comparison with the ternary operator
21    printf("The max value is: %d\n", ((a>b)?a:b));
22
23    return 0; // return an integer value of 0
24 }
```

```
os2024@os2024-virtualbox:~/Documents/lecture-2024-10-23$ gcc lec-04.c && ./a.out
You are not an adult!
The max value is: 11
The max value is: 11
os2024@os2024-virtualbox:~/Documents/lecture-2024-10-23$
```

2) CONTROL STRUCTURES: SWITCH

```
1 #include <stdio.h>
2
3 int main() {
4     int numberOfComputers=0;
5     printf("How many computers do you have at home?");
6     scanf("%d", &numberOfComputers);
7
8     switch(numberOfComputers) {
9         case 0:
10             printf("It is good for the environment!\n");
11             break;
12         case 1:
13             printf("You can code at home!\n");
14             break;
15         case 2:
16             printf("It is good if one of them is not working.\n");
17             break;
18         case 3:
19             printf("It is a lot of computers.\n");
20             break;
21         case 4:
22         case 5:
23         case 6:
24             printf("You should recycle several of them. :)\n");
25             break;
26         default:
27             printf("Wow, so many computers!\n");
28     }
29
30     return 0;
31 }
```

```
os2025@os2025-virtualbox:~/c-code$ gcc switch.c
os2025@os2025-virtualbox:~/c-code$ ./a.out
How many computers do you have at home?0
It is good for the environment!
os2025@os2025-virtualbox:~/c-code$ ./a.out
How many computers do you have at home?1
You can code at home
os2025@os2025-virtualbox:~/c-code$ ./a.out
How many computers do you have at home?4
You should recycle several of them. :)
os2025@os2025-virtualbox:~/c-code$ ./a.out
How many computers do you have at home?10
Wow, so many computers!
os2025@os2025-virtualbox:~/c-code$
```

3) LOOPS: WHILE, FOR

```
1 #include <stdio.h> // include the library for printf()
2 int sumOfIntegers(int start, int numberOfIntegers); // prototype of the function
3 int main() {
4     int i=0;
5     while (i<100) {
6         printf("%d/", i);
7         i++;
8     }
9     printf("\n");
10    for (int i=0; i<100;i++) {
11        printf("%d:", i);
12    }
13    printf("\n");
14    int sum=0; // summation in math
15    for (int i=0; i<100;i++) {
16        sum+=i;
17    }
18    printf("Sum: %d\n", sum);
19    int a=10, b=100;
20    printf("The sum of the %d first integers starting at %d is: %d\n", b, a, sumOfIntegers(a, b));
21    a=123, b=10;
22    printf("The sum of the %d first integers starting at %d is: %d\n", b, a, sumOfIntegers(a, b));
23    a=100, b=10;
24    printf("The sum of the %d first integers starting at %d is: %d\n", b, a, sumOfIntegers(a, b));
25    return 0; // return an integer value of 0
26 }
27
28 int sumOfIntegers(int start, int numberOfIntegers) {
29     // summation in math
30     int sum=0;
31     for (int i=start; i<numberOfIntegers+start; i++) {
32         sum+=i;
33     }
34     printf("Sum: %d\n", sum);
35     return sum;
36 }
37 }
38
```


3) LOOPS: WHILE, FOR

```
os2024@os2024-virtualbox:~/Documents/lecture-2024-10-23$ gcc lec-05.c && ./a.out
0/1/2/3/4/5/6/7/8/9/10/11/12/13/14/15/16/17/18/19/20/21/22/23/24/25/26/27/28/29/30/31/32/33/34/35/36/37/38/39/40/41
/42/43/44/45/46/47/48/49/50/51/52/53/54/55/56/57/58/59/60/61/62/63/64/65/66/67/68/69/70/71/72/73/74/75/76/77/78/79/
80/81/82/83/84/85/86/87/88/89/90/91/92/93/94/95/96/97/98/99/
0:1:2:3:4:5:6:7:8:9:10:11:12:13:14:15:16:17:18:19:20:21:22:23:24:25:26:27:28:29:30:31:32:33:34:35:36:37:38:39:40:41
:42:43:44:45:46:47:48:49:50:51:52:53:54:55:56:57:58:59:60:61:62:63:64:65:66:67:68:69:70:71:72:73:74:75:76:77:78:79:
80:81:82:83:84:85:86:87:88:89:90:91:92:93:94:95:96:97:98:99:
Sum: 4950
Sum: 5950
The sum of the 100 first integers starting at 10 is: 5950
Sum: 1275
The sum of the 10 first integers starting at 123 is: 1275
Sum: 1045
The sum of the 10 first integers starting at 100 is: 1045
os2024@os2024-virtualbox:~/Documents/lecture-2024-10-23$
```

4) COMPLEX DATATYPES: STRUCT

```
1 #include <stdio.h>
2 #include <string.h>
3 #include "struct.h"
4
5 int main() {
6     // Declare and populate two struct person variables
7     struct person person1;
8     struct person person2;
9
10    // Populate person1
11    strcpy(person1.firstname, "John");
12    strcpy(person1.lastname, "Doe");
13    person1.age = 30;
14
15    // Populate person2
16    strcpy(person2.firstname, "Jane");
17    strcpy(person2.lastname, "Smith");
18    person2.age = 25;
19
20    // Print the information
21    printf("Person 1: %s %s, %d years old\n", person1.firstname, person1.lastname, person1.age);
22    printf("Person 2: %s %s, %d years old\n", person2.firstname, person2.lastname, person2.age);
23
24    return 0;
25 }
```

File struct.c

4) COMPLEX DATATYPES: STRUCT

```
1
2 // defines the struct person
3
4 struct person {
5     char firstname[50];
6     char lastname[50];
7     int age;
8 };
9
```

File struct.h

```
os2025@os2025-virtualbox:~/c-code$ gcc struct.c
os2025@os2025-virtualbox:~/c-code$ ./a.out
Person 1: John Doe, 30 years old
Person 2: Jane Smith, 25 years old
os2025@os2025-virtualbox:~/c-code$
```

4) COMPLEX DATATYPES: ARRAYS

```
1 #include <stdio.h>
2
3 int main() {
4     float temperatures[12];
5     float sum = 0.0;
6     float average;
7
8     // Prompt the user to enter temperatures for each month
9     printf("Enter the temperature for each month:\n");
10    for (int i = 0; i < 12; i++) {
11        printf("Month %d: ", i + 1);
12        scanf("%f", &temperatures[i]);
13        sum += temperatures[i];
14    }
15
16    // Calculate the average
17    average = sum / 12;
18
19    // Print the average
20    printf("\nThe average temperature over 12 months is: %.2f\n", average);
21
22    return 0;
23 }
24
```

4) COMPLEX DATATYPES: ARRAYS

```
os2025@os2025-virtualbox:~/c-code$ ./a.out
Enter the temperature for each month:
Month 1: -2
Month 2: -4
Month 3: 1
Month 4: 7
Month 5: 14
Month 6: 20
Month 7: 22
Month 8: 24
Month 9: 20
Month 10: 17
Month 11: 8
Month 12: 1

The average temperature over 12 months is: 10.67
os2025@os2025-virtualbox:~/c-code$
```

4) COMPLEX DATATYPES: POINTERS

```
1 #include <stdio.h>
2
3 int main() {
4     int * ad; // declares a pointer to an integer
5     int n;    // declares an integer
6     n=20;    // affects the value 20 to the integer
7     ad=&n;   // affects the address of the integer to the address ad
8     *ad=30;  // affects the value 30 to the integer pointed by address ad
9
10    printf("The value is: %d\n", n);
11    printf("The address of n is: %p\n", &n);
12    printf("The address value ad is: %p\n", ad);
13    printf("The address of the pointer ad is: %p\n", &ad);
14
15    n=40;
16
17    printf("The value pointed by ad is: %d\n", *ad);
18    printf("The size of a pointer to an int is: %ld\n", sizeof(int *));
19
20    return 0;
21 }
22
```

4) COMPLEX DATATYPES: POINTERS

```
os2025@os2025-virtualbox:~/c-code$ gcc pointers.c
os2025@os2025-virtualbox:~/c-code$ ./a.out
The value is: 30
The address of n is: 0x7ffffd04d0cc
The address value ad is: 0x7ffffd04d0cc
The address of the pointer ad is: 0x7ffffd04d0d0
The value pointed by ad is: 40
The size of a pointer to an int is: 8
os2025@os2025-virtualbox:~/c-code$
```

4) FUNCTIONS

```
int sumOfIntegers(int start, int numberOfIntegers) {  
    // summation in math  
    int sum=0;  
    for (int i=start; i<numberOfIntegers+start; i++) {  
        sum+=i;  
    }  
    printf("Sum: %d\n", sum);  
  
    return sum;  
}
```


ANY QUESTIONS?